

Notes on Vakil's *The Rising Sea*

Foundations of Algebraic Geometry

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Reading log

- **May 2026** — started; setting up notation and motivation.

1 Introduction and motivation

What is the book trying to do? Replace the classical varieties-over-a-field treatment with the Grothendieck-school foundation: schemes, sheaves of modules, cohomology. The point is to build the vocabulary in which modern algebraic geometry is actually written.

Vakil's stylistic bet. Develop the abstract machinery first and let the geometric intuition arrive via examples that are forced through the machinery, rather than the other way around. “The rising sea” is Grothendieck’s metaphor: do not chisel through the rock, flood it.

What I want from this pass

- Comfort with the language: schemes, morphisms, quasi-coherent sheaves, cohomology of sheaves.
- Worked exercises through at least the schemes / quasi-coherent sheaves chapters.
- Enough fluency to read a paper that uses this language without translation.

2 Category theory background (Ch. 1)

Notes go here.

2.1 Categories, functors, natural transformations

3 Sheaves (Ch. 2)

Notes go here.

4 Toward affine schemes (Ch. 3)

Notes go here.

5 Worked exercises

Numbered to match Vakil's numbering. Add as worked.

Open questions / things to come back to

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